Vector Calculus and Complex Analysis Classwork (2025/2/20)

Name:

Student ID:

1. Let $f(x,y,z) = xy^2z^3$. Find an unit vector $\boldsymbol{u}$ in $\mathbb{R}^3$ such that $\left|D_{\boldsymbol{u}} f(1,1,1)\right|$ is maximum. Hence calculate $D_{\boldsymbol{u}} f(1,1,1)$.

2. Find the value of the line integral $\int_C \boldsymbol{F}(\boldsymbol{r}) \cdot d\boldsymbol{r}$ when $\boldsymbol{F}(\boldsymbol{r}) = (xy, yz, zx)$ and the arc C is parametrized by $\boldsymbol{r}(t) = (t, t^2, t^3)$ from (1,1,1) to (2,4,8).

Solution

1. $\nabla f(1,1,1) = (y^2z^3, 2xyz^3, 3xy^2z^2)_{(1,1,1)} = (1,2,3)$

$$\boldsymbol{u} = \frac{1}{\sqrt{1^2+2^2+3^2}}(1,2,3) = \frac{1}{\sqrt{14}}(1,2,3)$$

$$D_{\boldsymbol{u}} f(1,1,1) = \frac{1}{\sqrt{14}}(1,2,3)\cdot(1,2,3) = \sqrt{14}$$

2. $d\boldsymbol{r} = (1,2t,3t^2)dt,\ \boldsymbol{F}(\boldsymbol{r}(t)) = (t^3,t^5,t^4).$

$$\int_C \boldsymbol{F}(\boldsymbol{r})\cdot d\boldsymbol{r} = \int_1^2 (t^3,t^5,t^4)\cdot(1,2t,3t^2)dt = \int_1^2 (t^3+3t^6)dt = \left[\frac{t^4}{4}+\frac{3t^7}{7}\right]_1^2 = \frac{2^4-1}{4}+\frac{3(2^7-1)}{7}$$